Syllabus Edition
First teaching 2025
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The Use of Technology in the Production of Goods & Services

Exam code: 0450, 0986 & 0264, 0774

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Written by: Lisa Eades
Reviewed by: Steve Vorster
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The impact of technology on production methods

- Technology has changed how businesses produce goods and services
- It can make production faster, cheaper and more accurate, but also comes with some challenges

Advantages and disadvantages of technology in production

Advantages	Disadvantages

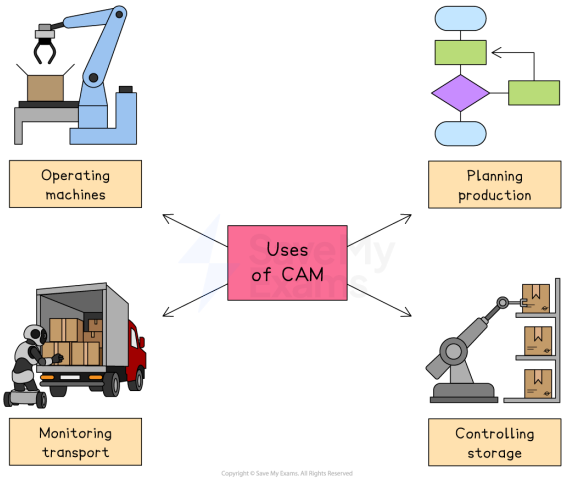
• **Increased speed and efficiency**
  ○ Machines and automation can produce goods faster than humans, especially in **flow production**
• **Lower labour costs**
  ○ Automation reduces the need for large numbers of workers, especially unskilled labour
• **Greater accuracy and consistency**
  ○ Technology improves product
	• **High initial costs**
	  ○ Buying and installing new machinery or software can be expensive for small businesses
	• **Training required**
	  ○ Workers may need new skills to operate machines or software, which takes time and money
	• **Risk of unemployment**
	  ○ Replacing workers with machines may lead to job losses,

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

○ Modern technology allows quick changes between batches or designs	• **Dependence on technology** ○ A breakdown in machines or systems can stop production entirely and cause delays

Computer-aided manufacture (CAM)

- Computer-Aided Manufacture (CAM) is the use of computers to control and assist with production activities



CAM can be used to operate machines, plan production, monitor transport and control storage

Uses of computer-aided manufacture

Operating machines

- CAM controls machinery such as cutters, drills or robotic arms, ensuring tasks are completed quickly, accurately, and with consistent quality

Planning production schedules

- CAM software helps plan when and how products will be made, reducing delays and making sure resources are used efficiently

Monitoring transport

- CAM systems can track the movement of materials and products, helping ensure they reach the right stage of production on time

Controlling storage

3D printing

- Digital designs are uploaded and the 3D printer produces the product **with minimal labour costs**
- Digital printers have been used to print car engines, houses, food and replacement limbs

The impact of 3D printing on production

Impact	Explanation
Cost	• Traditional methods need expensive tooling and machinery, which is a high upfront cost for small businesses • 3D printing reduces costs as no special tools are required
Productivity	• Traditional production has long lead times due to set-up and tooling, causing delays • 3D printing is faster with minimal lead times, allowing quicker response to customer demand and faster product launches
Quality	• 3D printing produces detailed, high-quality items with greater accuracy • Traditional methods may struggle with complex designs
Flexibility	• 3D printing allows small, low-cost production runs tailored to specific customer needs • Traditional production is less suited to customisation

Examiner Tips and Tricks

Examiners value awareness that technology is **constantly evolving** – mention that businesses must keep up with new developments to stay competitive and avoid falling behind rivals

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